

David Manuel

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years of backend and data engineering experience, specializing in Python-driven enterprise architectures, high-performance data platforms, and scalable investment data infrastructure. Expertise in designing RESTful APIs, optimizing complex SQL and NoSQL databases, and building robust data integration pipelines using Airflow, Kafka, Docker, and Redis. Adept at leveraging SOLID principles, Domain-Driven Design (DDD), and automated acceptance testing to build reliable, high-availability services in dynamic Agile environments. Seeking a remote Senior Python Engineer position.

CORE COMPETENCIES

Languages: Python (Primary/Open Source), SQL, C#, JavaScript, TypeScript, Java

Data Engineering & Orchestration: Apache Airflow, Apache Kafka, RabbitMQ, Redis, Data Pipelines, ETL/ELT

Databases: PostgreSQL, SQL Server, MySQL, MongoDB, NoSQL, Pinecone (Vector DB)

APIs & Architecture: RESTful APIs, GraphQL, Microservices, Domain-Driven Design (DDD), SOLID Principles, System Architecture

Cloud, DevOps & CI/CD: Docker, Kubernetes, Ansible, Terraform, AWS, Google Cloud Platform, GitHub Actions, Azure DevOps

Security & Testing: OAuth2, JWT, Enterprise SSO, RBAC, PlainID, Okta, Cybersecurity Best Practices, Unit Testing, Integration Testing, Automated Acceptance Testing

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Dec 2022 – Present | Rochester, NY

Netsmartz

- Built scalable enterprise application services and high-performance Python backend microservices powering an enterprise data platform and AI workflows.
- Cut median API response time from 850ms to 380ms (55% improvement) by redesigning data ingestion layers, implementing Redis caching strategies, and optimizing PostgreSQL and Elasticsearch queries.
- Architected robust ETL/ELT data transfer pipelines capable of handling high-volume datasets within strict SLAs across PostgreSQL, MongoDB, Redis, and vector search systems.
- Integrated secure identity and access control frameworks utilizing OAuth2, JWT, Okta, and Role-Based Access Control (RBAC) to mitigate cybersecurity risks across enterprise endpoints.
- Built and enhanced CI/CD automation pipelines utilizing Docker, Kubernetes, and Terraform on AWS (EKS, Lambda) while enforcing SOLID principles and test-driven standards.
- Led code and design reviews, maintained technical wiki documentation, and mentored a 5-person engineering team in an Agile delivery model.

Senior Full Stack Software Engineer

Feb 2020 – Nov 2022 | Chicago, IL

Tempus AI

- Designed and deployed secure RESTful APIs and asynchronous data services using Python (FastAPI/Flask) and .NET Core to support high-availability analytics platforms.

- Reduced p95 API latency from 620ms to 470ms (24% improvement) through SQL query tuning, schema indexing, and database connection pooling across large-scale relational databases.
- Engineered secure, enterprise-grade authentication mechanisms enforcing OAuth2, JWT, field-level encryption, and comprehensive audit logging for sensitive healthcare and analytics datasets.
- Containerized core backend services with Docker and orchestrated cloud deployments across AWS and GCP using automated acceptance testing and CI/CD best practices.
- Formulated scalable data storage and validation routines using SQL and NoSQL storage layers to maintain high data quality control and consistent refresh rates.

Software Engineer

Dec 2015 – Jan 2020

Talmate

- Developed high-throughput backend services and RESTful APIs, cutting workflow execution time from 12 minutes to under 6 minutes (50% reduction) via Redis caching and backend optimization.
- Built automated data integration scripts and web services in Python and Java to streamline third-party system connections and data pipeline workflows.
- Automated CI/CD build and deployment pipelines using GitHub Actions, Azure DevOps, and Docker containers, accelerating time-to-production for core modules.
- Collaborated with business analysts, product owners, and cross-functional teams to define user stories, story estimates, and delivery roadmaps in an Agile environment.

EDUCATION

Bachelor's Degree in Computer Science

Aug 2011 – May 2015 | Houston, TX

University of Houston